Regional Dialect Synthesis Pipeline

Milestone 3: Model Architecture & End-to-End Pipeline

Standard Hindi → Haryanvi (Bangru) Dialect

Deadline: 19 March 2026

Scroll down to navigate slides ↓

Team Contributions (Worklog)

Milestone 3 Objective: Design, justification, and end-to-end pipeline verification.

Abhishek

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- Designed Hindi → Haryanvi translation pipeline using LLaMA 3.1 + QLoRA
- Implemented instruction-based formatting and tokenizer integration
- Ran inference on subset data and validated outputs
- Ensured output compatibility for TTS module

Satyam Srivastava

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- Created Milestone-3 report (model architecture + pipeline explanation)
- Updated worklog.md and managed final submission

Sanket Agrawal

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- Verified dataset consistency with model input requirements
- Uploaded processed datasets to Hugging Face
- Assisted in pipeline testing and validation
- Prepared initial PPT structure for milestone presentation

Md Fazlur Rahman

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- Designed TTS architecture (VITS-based)
- Defined audio preprocessing pipeline
- Structured dataset into TTS-ready format

Scroll down to navigate slides ↓

1. Pipeline Overview & Data Organization

The system bridges the low-resource dialect barrier by splitting the task into two sequential domains to prevent memory overhead and data leakage.

Module A: Text-to-Text

- **Task:** Standard Hindi → Haryanvi
- **Dataset:** `Satyam-Srivastava/RDS` (Parallel corpus)
- **Split:** Train (80%), Val (10%), Test (10%)

Module B: Text-to-Audio

- **Task:** Haryanvi Text → Speech
- **Dataset:** `ankitdhiman/haryanvi-tts` (~4k pairs)
- **Structure:** LJSpeech format (wavs + metadata.csv)

Scroll down to navigate slides ↓

2. Preprocessing & Formatting

Text Pipeline (LLaMA 3.1)

- **Length Filtering:** Sentences constrained to 3–25 words to prevent padding inefficiencies.
- **Template Formatting:** Mapped to Instruction format.
  System: "You are a regional dialect translator..."
  User: "तुम कहाँ जा रहे हो?" → *Assistant:* "तू कड़े जा रया सै?"
- **Tokenization:** Padding via EOS token; max sequence length set to 256.

Audio Pipeline (VITS)

- **Extraction:** Bypassed HuggingFace `torchcodec` bug by streaming raw bytes directly via `soundfile`.
- **Resampling:** Enforced strict standardization to **22050 Hz**.
- **Phonemization:** Applied `espeak-ng (hi)` to convert Devanagari to IPA, handling critical North Indian schwa-deletion.

Scroll down to navigate slides ↓

3. Model Architectures & Justification

Module A: LLaMA 3.1 (8B) + QLoRA

Why this choice?

- Traditional Seq2Seq (like T5) fails due to the lack of millions of parallel sentences.
- LLaMA 3.1 possesses deep semantic knowledge of Hindi. LoRA adapters (4-bit NF4 quantized) efficiently "steer" this knowledge toward Bangru syntax.

Module B: VITS (Transfer Learning)

Why this choice?

- **Stochastic Duration Predictor:** Captures Haryanvi's aggressive syllabic compression better than FastSpeech2's rigid metronomic cadence.
- **Transfer Learning:** Initialized with AI4Bharat `vits_rasa_13` Hindi checkpoints to solve the cold-start problem on a small 4,000-sample dataset.

Scroll down to navigate slides ↓

4. End-to-End Data Flow

```
                    [RAW HINDI INPUT] ("मैं आज बाजार जा रहा हूँ")
                                  |
                                  ▼
(LLaMA Tokenizer) ————> [Input_IDs: Shape (Batch, Seq_Len)]
                                  |
                                  ▼
 [Module A: LLaMA 3.1 + QLoRA] —> (Cross-Entropy Prediction)
                                  |
                                  ▼
                [HARYANVI TEXT OUTPUT] ("मैं आज बाजार जा रया सै।")
                                  |
                                  ▼
(espeak-ng Phonemizer) —> [Phoneme Sequence: Shape (Batch, Phoneme_Len)]
                                  |
                                  ▼
                [Module B: VITS Encoder & Duration Predictor]
                                  |
                                  ▼
        [Latent Acoustic Space] —> (Flow-based Decoder + HiFi-GAN)
                                  |
                                  ▼
                  [RAW AUDIO WAVEFORM] (.wav file @ 22050Hz)
```


Scroll down to navigate slides ↓

5. Pipeline Verification (Tracer Bullet)

We executed a highly restricted subset run to verify mathematical integration, dimension matching, and end-to-end functionality.

Component	Execution Result	Status
LLaMA + QLoRA	Loss decreased from 0.094 to 0.090 over 3 epochs. Adapters merged successfully.	**Passed**
Text Output	"यह किताब मेरी है" → "या किताब मेरी सै"	**Passed**
VITS Audio Pipeline	Streamed 50 samples. GAN loops stabilized. Output `.wav` generated from translated text.	**Passed**

Scroll down to navigate slides ↓

6. Conclusion & Next Steps

This milestone successfully validates the structural integrity of the project. Data flows perfectly from raw Standard Hindi text to a Haryanvi audio waveform without shape mismatches or dataset loading bottlenecks.

Metrics Established for Final Training:

- **Text Module:** Causal LM Loss, chrF++ (optimal for morphologically rich dialects), BLEU Score.
- **Audio Module:** Mel-spectrogram L1 Loss, KL-Divergence, Mean Opinion Score (MOS), Mel-Cepstral Distortion (MCD).

Milestone 4 Objectives:

- Initiate full-scale TTS training on all 4,000 samples.
- Implement strategic layer freezing for the VITS Text Encoder.
- Generate quantitative benchmark metrics.

Scroll down to navigate slides ↓